Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*			**Indicative content:** **Changing temperature** 1. Along AB the temperature of the ice increases and the vibrational /kinetic energy of the molecules increases. 2. Along CD the temperature of water increases and the KE of molecules increases. 3. Along EF the steam temperature increases and KE of molecules increases. **Changing state from solid to liquid** 4. Along BC the energy supplied breaks bonds between molecules 5. ......which allows them to move past one another. **Changing state from liquid to gas** 6. Along DE the energy supplied breaks bonds between molecules 7. ....which allows them to escape from one another. 8. More bonds broken when boiling so DE longer than BC. **5 – 6 marks** A detailed description including all three sections *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3 – 4 marks** A partial description which may be from 2 or 3 sections. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.*	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Any relevant comment on any aspect of the graph *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
	(b)	(i)		It takes 336 000 [J] to melt 1 kg of ice (1) …into water at the same temperature / at its melting point (1)	2			2		
		(ii)		Conversion 800 g to 0.8 kg (1) Substitution with correct temperature change into $\Delta Q = mc\Delta\theta$ i.e. 0.8 × 2 030 × 23 (1) = 37 352 [J] (1) Substitution and answer into $Q = mL$ i.e. 0.8 × 336 000 = 268 800 [J] (1) Total energy = **306 152** ecf [J] (1) → 5 marks 306 152 000 J → 4 marks 37 352 J →3 marks 37 352 000 J → 2 marks 268 800 J → 2 marks 268 800 000 J → 1 mark		5		5	5	
				Question 6 total	**8**	**5**	**0**	**13**	**5**	**0**